

Problems on H.C.F and L.C.M

1. Find the H.C.F of 144, 336 and 2016.

- A) 24
- B) 36
- C) 48
- D) 56

Answer: C) 48

2. Find the L.C.M of 12, 15, 20 and 27.

- A) 480
- B) 540
- C) 620
- D) 720

Answer: B) 540

3. Two numbers are in the ratio 4:5. If their H.C.F is 16, then the sum of these numbers is:

- A) 124
- B) 144
- C) 154
- D) 164

Answer: B) 144

4. Find the H.C.F of 204, 1190 and 1445.

- A) 17
- B) 19
- C) 21
- D) 23

Answer: A) 17

5. The H.C.F of two numbers is 18 and their L.C.M is 378. If one of the numbers is 54, then the other number is:

- A) 126
- B) 144
- C) 198
- D) 238

Answer: A) 126

6. What is the greatest number that will divide 307 and 330 leaving remainders 3 and 7 respectively?

- A) 16
- B) 17
- C) 19
- D) 23

Answer: C) 19

7. The L.C.M of two numbers is 2310 and their H.C.F is 30. If one of the numbers is 210, the other is:

- A) 330
- B) 1470
- C) 2100
- D) 16170

Answer: A) 330

8. Find the least number which is exactly divisible by 12, 15, 20 and 27.

- A) 450
- B) 520
- C) 540
- D) 600

Answer: C) 540

9. The product of two numbers is 1600 and their H.C.F is 5. The L.C.M of the numbers is:

- A) 8000

B) 1600

C) 320

D) 1605

Answer: C) 320

10. Find the greatest number which divides 285 and 1249 leaving remainders 9 and 7 respectively.

A) 46

B) 69

C) 138

D) 276

Answer: C) 138

11. The least number, which when divided by 12, 15, 20 and 54 leaves in each case a remainder of 8, is:

A) 504

B) 536

C) 544

D) 548

Answer: D) 548

12. The H.C.F of two numbers is 11 and their L.C.M is 7700. If one of the numbers is 275, the other is:

A) 279

B) 283

C) 308

D) 318

Answer: C) 308

13. Three numbers are in the ratio of 3:4:5 and their L.C.M is 2400. Their H.C.F is:

A) 40

B) 80

C) 120

D) 200

Answer: A) 40

14. Find the H.C.F of $\frac{3}{4}$, $\frac{5}{6}$ and $\frac{6}{7}$.

A) $\frac{1}{84}$

B) $\frac{1}{54}$

C) $\frac{1}{504}$

D) $\frac{1}{24}$

Answer: A) $\frac{1}{84}$

15. Find the L.C.M of $\frac{2}{3}$, $\frac{3}{5}$, $\frac{4}{7}$ and $\frac{9}{13}$.

A) 36

B) $\frac{1}{36}$

C) $\frac{1}{1365}$

D) $\frac{12}{455}$

Answer: A) 36

16. The H.C.F of two numbers is 15 and their product is 6300. The number of possible pairs of numbers is:

A) 4

B) 3

C) 2

D) 1

Answer: C) 2

17. The L.C.M of two numbers is 48. The numbers are in the ratio 2:3. The sum of the numbers is:

A) 28

B) 32

C) 40

D) 64

Answer: C) 40

18. What is the smallest number which when diminished by 7, is divisible by 12, 16, 18, 21 and 28?

- A) 1008
- B) 1015
- C) 1022
- D) 1032

Answer: B) 1015

19. The H.C.F of two numbers is 8. Which of the following can never be their L.C.M?

- A) 24
- B) 48
- C) 56
- D) 60

Answer: D) 60

20. The product of two co-primes is 117. Their L.C.M should be:

- A) 1
- B) 117
- C) equal to their H.C.F
- D) cannot be calculated

Answer: B) 117

21. The sum of two numbers is 528 and their H.C.F is 33. The number of pairs of numbers satisfying the above condition is:

- A) 4
- B) 6
- C) 8
- D) 12

Answer: A) 4

22. Find the largest number of 4 digits divisible by 12, 15, 18 and 27.

A) 9690

B) 9720

C) 9930

D) 9960

Answer: B) 9720

23. Find the least multiple of 23, which when divided by 18, 21 and 24 leaves 7, 10 and 13 as remainders respectively.

A) 3002

B) 3013

C) 3024

D) 3036

Answer: B) 3013

24. The H.C.F of two numbers is 12 and their difference is 12. The numbers are:

A) 66, 78

B) 70, 82

C) 94, 106

D) 84, 96

Answer: D) 84, 96

25. A, B and C start at the same time in the same direction to run around a circular stadium. A completes a round in 252 seconds, B in 308 seconds and C in 198 seconds, all starting at the same point. After what time will they meet again at the starting point?

A) 26 min 18 sec

B) 42 min 36 sec

C) 45 min

D) 46 min 12 sec

Answer: D) 46 min 12 sec

26. Find the greatest number which will divide 410, 751 and 1030 so as to leave the remainder 7 in each case.

A) 31

B) 34

C) 37

D) 41

Answer: A) 31

27. Six bells commence tolling together and toll at intervals of 2, 4, 6, 8, 10 and 12 seconds respectively. In 30 minutes, how many times do they toll together?

A) 4

B) 10

C) 15

D) 16

Answer: D) 16

28. The H.C.F of 0.63, 1.05 and 2.1 is:

A) 0.021

B) 0.21

C) 2.1

D) 21

Answer: B) 0.21

29. Find the L.C.M of 3, 2.7 and 0.09.

A) 2.7

B) 27

C) 0.27

D) 2700

Answer: B) 27

30. The greatest number that exactly divides 105, 1001 and 2436 is:

A) 3

B) 7

C) 11

D) 21

Answer: B) 7

31. If the sum of two numbers is 55 and the H.C.F and L.C.M of these numbers are 5 and 120 respectively, then the sum of the reciprocals of the numbers is equal to:

A) $55/601$

B) $601/55$

C) $11/120$

D) $120/11$

Answer: C) $11/120$

32. Find the smallest number which leaves 22, 35 and 48 as remainders when divided by 35, 48 and 61 respectively.

A) 10244

B) 10257

C) 10270

D) 10231

Answer: B) 10257

33. Two numbers are in the ratio 3:4. Their L.C.M is 84. The greater number is:

A) 21

B) 24

C) 28

D) 84

Answer: C) 28

34. The H.C.F of two numbers is 23 and the other two factors of their L.C.M are 13 and 14. The larger of the two numbers is:

A) 276

B) 299

C) 322

D) 345

Answer: C) 322

35. Determine the H.C.F of $2^3 \times 3^2 \times 5$ and $2^2 \times 3^3 \times 5^2$.

A) 360

B) 180

C) 90

D) 45

Answer: B) 180

36. Find the side of the largest possible square slabs which can be paved on the floor of a room 2m 50cm long and 1m 50cm broad.

A) 40 cm

B) 50 cm

C) 55 cm

D) 60 cm

Answer: B) 50 cm

37. The H.C.F of 2923 and 3239 is:

A) 37

B) 47

C) 73

D) 79

Answer: D) 79

38. Three different containers contain 496 liters, 403 liters and 713 liters of mixtures of milk and water respectively. What biggest measure can measure all the different quantities exactly?

A) 1 liter

B) 7 liters

C) 31 liters

D) 41 liters

Answer: C) 31 liters

39. Find the greatest number of four digits which is divisible by 15, 25, 40 and 75.

A) 9000

B) 9400

C) 9600

D) 9800

Answer: C) 9600

40. The least number which when divided by 5, 6, 7 and 8 leaves a remainder 3, but when divided by 9 leaves no remainder, is:

A) 1677

B) 1683

C) 2523

D) 3363

Answer: B) 1683

41. Find the L.C.M of 72, 108 and 2100.

A) 37800

B) 2100

C) 36600

D) 28400

Answer: A) 37800

42. The L.C.M of two numbers is 495 and their H.C.F is 5. If the sum of the numbers is 100, then their difference is:

A) 10

B) 46

C) 70

D) 90

Answer: A) 10

43. 105 goats, 140 donkeys and 175 cows have to be taken across a river. There is only one boat which will have to make many trips. The boatman insists that he will take the same number of animals in every trip and they have to be of the same kind. He will naturally like to take the largest possible number each time. Can you tell how many animals went in each trip?

A) 35

B) 70

C) 25

D) 15

Answer: A) 35

44. Which of the following has the most number of divisors?

A) 99

B) 101

C) 176

D) 182

Answer: C) 176

45. The L.C.M of two numbers is 20 times their H.C.F. The sum of H.C.F and L.C.M is 2520. If one of the numbers is 480, then the other number is:

A) 400

B) 480

C) 520

D) 600

Answer: D) 600

46. Three planets revolve round the Sun once in 200, 250 and 300 days respectively in their own orbits. When do they all come relatively to the same position as at a certain point of time in their orbits?

A) After 3000 days

B) After 2000 days

C) After 1500 days

D) After 1200 days

Answer: A) 3000 days

47. The H.C.F of two numbers is 12. Which one of the following can never be their L.C.M?

- A) 72
- B) 60
- C) 90
- D) 84

Answer: C) 90

48. H.C.F of 3240, 3600 and a third number is 36 and their L.C.M is $2^4 \times 3^5 \times 5^2 \times 7^2$. The third number is:

- A) $2^2 \times 3^5 \times 7^2$
- B) $2^2 \times 5^3 \times 7^2$
- C) $2^5 \times 5^2 \times 7^2$
- D) $2^3 \times 3^5 \times 7^2$

Answer: A) $2^2 \times 3^5 \times 7^2$

49. Three numbers are in the ratio 1:2:3 and their H.C.F is 12. The numbers are:

- A) 6, 12, 18
- B) 12, 24, 36
- C) 12, 24, 30
- D) 5, 10, 15

Answer: B) 12, 24, 36

50. The sum of two numbers is 2000 and their L.C.M is 21879. The two numbers are:

- A) 1993, 7
- B) 1991, 9
- C) 1989, 11
- D) 1987, 13

Answer: C) 1989, 11

51. Find the greatest number which on dividing 1657 and 2037 leaves remainders 6 and 5 respectively.

A) 123

B) 127

C) 235

D) 305

Answer: B) 127

52. The least multiple of 7, which leaves a remainder of 4, when divided by 6, 9, 15 and 18 is:

A) 74

B) 94

C) 184

D) 364

Answer: D) 364

53. The greatest number which can divide 284, 698 and 1618 leaving the same remainder 8 in each case is:

A) 46

B) 23

C) 92

D) 138

Answer: A) 46

54. The H.C.F and L.C.M of two numbers are 50 and 250 respectively. If the first number is divided by 2, the quotient is 50. The second number is:

A) 50

B) 100

C) 125

D) 250

Answer: C) 125

55. Find the largest number which divides 62, 132 and 237 to leave the same remainder in each case.

A) 8

B) 35

C) 57

D) 75

Answer: B) 35

56. The product of two numbers is 1280 and their H.C.F is 8. The L.C.M of the numbers will be:

A) 128

B) 160

C) 150

D) 140

Answer: B) 160

57. The least perfect square number divisible by 3, 4, 5, 6 and 8 is:

A) 900

B) 1200

C) 2500

D) 3600

Answer: D) 3600

58. What is the smallest number which leaves a remainder of 3 when divided by any of the numbers 5, 6 or 8 but leaves no remainder when it is divided by 9?

A) 123

B) 603

C) 723

D) 243

Answer: D) 243

59. Two numbers are in the ratio 3:5. Their L.C.M is 225. The smaller number is:

A) 45

B) 60

C) 75

D) 90

Answer: A) 45

60. The sum of two numbers is 36 and their H.C.F is 4. How many pairs of such numbers are possible?

- A) 1
- B) 2
- C) 3
- D) 4

Answer: C) 3

61. The least number which when divided by 48, 60, 72, 108 and 140 leaves 38, 50, 62, 98 and 130 as remainders respectively is:

- A) 11115
- B) 15110
- C) 15120
- D) 15210

Answer: B) 15110

62. Find the greatest number that will divide 964, 1238 and 1400 leaving remainders of 41, 31 and 51 respectively.

- A) 61
- B) 71
- C) 73
- D) 81

Answer: B) 71

63. Three numbers are in the ratio 2:3:4. The sum of their squares is 1856. The numbers are:

- A) 8, 12, 16
- B) 16, 24, 32
- C) 12, 18, 24
- D) None of these

Answer: B) 16, 24, 32

64. The product of two numbers is 1320 and their H.C.F is 6. The L.C.M of the numbers is:

- A) 220
- B) 1314
- C) 1326
- D) 7920

Answer: A) 220

65. 252 can be expressed as a product of primes as:

- A) $2 \times 2 \times 3 \times 3 \times 7$
- B) $2 \times 2 \times 2 \times 3 \times 7$
- C) $3 \times 3 \times 3 \times 3 \times 7$
- D) $2 \times 3 \times 3 \times 3 \times 7$

Answer: A) $2 \times 2 \times 3 \times 3 \times 7$

66. Find the lowest common multiple of 24, 36 and 40.

- A) 120
- B) 240
- C) 360
- D) 480

Answer: C) 360

67. The least number of five digits which is exactly divisible by 12, 15 and 18 is:

- A) 10010
- B) 10015
- C) 10020
- D) 10080

Answer: D) 10080

68. The ratio of two numbers is 4:5 and their L.C.M is 120. The numbers are:

- A) 30, 40

B) 24, 30

C) 35, 40

D) 40, 50

Answer: B) 24, 30

69. Find the L.C.M of 148 and 185.

A) 680

B) 740

C) 2960

D) 3700

Answer: B) 740

70. The sum of two numbers is 84 and their H.C.F is 12. Total number of such pairs of numbers is:

A) 2

B) 3

C) 4

D) 5

Answer: B) 3

71. Find the least number which is a multiple of 31 and when divided by 15, 24 and 32 leaves the remainders 2, 11 and 19 respectively.

A) 2356

B) 2387

C) 2325

D) 2418

Answer: B) 2387

72. H.C.F of two numbers is 13. If these two numbers are in the ratio of 15:11, then find the numbers.

A) 230, 140

B) 215, 130

C) 195, 143

D) 155, 115

Answer: C) 195, 143

73. Find the greatest number that will divide 148, 246 and 623 leaving remainders 4, 6 and 11 respectively.

A) 12

B) 10

C) 16

D) 18

Answer: A) 12

74. Two numbers are in the ratio 3:5 and their L.C.M is 75. The H.C.F is:

A) 3

B) 5

C) 10

D) 15

Answer: B) 5

75. Find the side of the largest possible square slabs which can be paved on the floor of a room 5m 44cm long and 3m 74cm broad.

A) 34 cm

B) 44 cm

C) 56 cm

D) 64 cm

Answer: A) 34 cm

76. A rectangular courtyard 3.78 meters long and 5.25 meters wide is to be paved exactly with square tiles, all of the same size. What is the largest size of the tile which could be used for the purpose?

A) 14 cm

B) 21 cm

C) 42 cm

D) 28 cm

Answer: B) 21 cm

77. The product of two numbers is 4107. If the H.C.F of these numbers is 37, then the greater number is:

A) 101

B) 107

C) 111

D) 185

Answer: C) 111

78. The least number which when increased by 5 is divisible by each one of 24, 32, 36 and 54 is:

A) 427

B) 859

C) 869

D) 4320

Answer: B) 859

79. Find the largest number which divides 25, 73 and 97 to leave the same remainder in each case.

A) 24

B) 21

C) 23

D) 6

Answer: A) 24

80. The greatest possible length which can be used to measure exactly the lengths 7 m, 3 m 85 cm, 12 m 95 cm is:

A) 15 cm

B) 25 cm

C) 35 cm

D) 42 cm

Answer: C) 35 cm

81. If the sum of two numbers is 36 and their H.C.F. and L.C.M. are 3 and 105 respectively, the sum of the reciprocals of the two numbers is:

A) $\frac{2}{35}$

B) $\frac{3}{35}$

C) $\frac{4}{35}$

D) $\frac{2}{25}$

Answer: C) $\frac{4}{35}$

82. Find the H.C.F of 513, 1134 and 1215.

A) 18

B) 27

C) 33

D) 36

Answer: B) 27

83. The L.C.M of two numbers is 45 times their H.C.F. If one of the numbers is 125 and the sum of H.C.F and L.C.M is 1150, the other number is:

A) 215

B) 220

C) 225

D) 235

Answer: C) 225

84. What is the greatest number which can be used to measure exactly the following lengths: 20 ft, 13 ft 9 inches, 17 ft 6 inches, 21 ft 3 inches?

A) 9 inches

B) 1 ft 3 inches

C) 1 ft 6 inches

D) 2 ft

Answer: B) 1 ft 3 inches

85. Find the H.C.F of 2923, 3239.

A) 37

B) 73

C) 79

D) 47

Answer: C) 79

86. Find the L.C.M of 0.6, 9.6, 0.36.

A) 9.6

B) 14.4

C) 28.8

D) 36

Answer: C) 28.8

87. Find the H.C.F of 1.75, 5.6 and 7.

A) 0.07

B) 0.7

C) 3.5

D) 0.35

Answer: D) 0.35

88. The H.C.F of 108, 288 and 360 is:

A) 18

B) 36

C) 54

D) 72

Answer: B) 36

89. 3 different electronic devices beep after every 48 seconds, 72 seconds and 100 seconds respectively. If they beeped together at 10 a.m., when will they beep together again?

A) 11:00 a.m.

B) 11:30 a.m.

C) 12:00 noon

D) 1:00 p.m.

Answer: A) 11:00 a.m.

90. Rectangular field is 320m by 200m. It is to be partitioned into equal square plots. What is the maximum possible area of such a square plot?

A) 1600 sq m

B) 1200 sq m

C) 800 sq m

D) 400 sq m

Answer: A) 1600 sq m

91. The product of H.C.F and L.C.M of two numbers is 24. The difference of two numbers is 2. Find the numbers.

A) 2, 4

B) 6, 4

C) 8, 6

D) 8, 10

Answer: B) 6, 4

92. The ratio of two numbers is 3:4. The product of their H.C.F and L.C.M is 2028. The sum of the numbers is:

A) 86

B) 91

C) 84

D) 96

Answer: B) 91

93. The H.C.F of x and y is z. Then their L.C.M is:

A) xy/z

B) xyz

C) xz/y

D) yz/x

Answer: A) xy/z

94. Find the greatest number which will divide 2112 and 2792 leaving the remainder 4 in each case.

A) 63

B) 68

C) 72

D) 78

Answer: B) 68

95. The L.C.M of three different numbers is 120. Which of the following cannot be their H.C.F?

A) 8

B) 12

C) 24

D) 35

Answer: D) 35

96. Find the L.C.M of 22, 54, 108 and 135.

A) 5940

B) 5490

C) 5640

D) 5840

Answer: A) 5940

97. The H.C.F of two numbers is 12906 and 14818 is 478. Their L.C.M is:

A) 400086

B) 200043

C) 600129

D) 800172

Answer: A) 400086

98. The greatest number of 5 digits divisible by 16, 24, 36 and 54 is:

A) 99360

B) 99792

C) 99640

D) 99936

Answer: B) 99792

99. The ratio of two numbers is 4:5 and their H.C.F is 8. The L.C.M is:

A) 120

B) 160

C) 240

D) 320

Answer: B) 160

100. If the product of two numbers is 324 and their H.C.F is 3, then their L.C.M is:

A) 98

B) 108

C) 112

D) 118

Answer: B) 108